Ryan Swift

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AI & AGENTIC SYSTEMS EXPERTISE

- **Agentic Systems**: Agent harness development, workflow orchestration, Model Context Protocol (MCP), tool integration, knowledge-base routing
- **LLM Evaluation & Safety**: Behavioral evaluation, prompt-injection testing, sycophancy, agent alignment
- **Representation Learning & Interpretability**: Knowledge distillation, mechanistic interpretability, applications of interpretability

EDUCATION

University of Southern California May 2029 (Expected)
PhD in Computer Science *Advisor: Konstantinos Psounis*

- Researching multimodal representation learning and its applications
- Interests include world models, interpretability, safety, and alignment
- Relevant Coursework: Trustworthy Foundation Models, High Dimensional Mathematics, Advanced NLP

University of California, Davis Jun 2024
B.S. in Computer Science *Advisors: Zubair Shafiq, Chen-Nee Chuah*

- Undergraduate researcher in the University Honors Program
- Designed a novel ML authentication pipeline, achieving 90% accuracy on user data
- Relevant Coursework: Optimization, Computer Security, Software Engineering, Theory of Computation

WORK EXPERIENCE

Graduate Student Intern at Lawrence Livermore National Laboratory May–Aug 2026
Mentor: Colin Ponce

- Co-developed a custom agent harness for integrating agentic workflows into critical infrastructure, including power systems, as part of the *Stormbreaker* project
- Designed and implemented the harness's Model Context Protocol (MCP) interface to support agent access to external and internal tools and data sources
- Added 8 tools to the harness, including agent access to threat intelligence databases
- Implemented workflow routing that required agents to consult internal knowledge bases before task execution
- Conducted independent research on agent safety and alignment, evaluating open-source LLM susceptibility to prompt injections designed to induce sycophantic responses

RESEARCH EXPERIENCE

Follow the Features Jan–Oct 2026
Collaborators: Konstantinos Psounis

- Developed an interpretable data-selection pipeline using sparse autoencoder (SAE) features to rank training examples and construct compact instruction-tuning datasets
- Designed feature scoring and diversity resampling, with a filtering pipeline reusable across target model families
- Evaluated LoRA fine-tuning across Llama and Qwen models using benchmark suites, local LLM-as-a-judge evaluation, and three-seed experiments
- Fine-tuned Qwen 3 8B on 100 selected examples from 50K+ Alpaca samples, exceeding full-dataset training by 1.59 percentage points on a combined instruction-following and benchmark metric

RAPID: Increasing Efficiency of Internal Distillation via Randomized Methods Aug–Sep 2026
Collaborators: Konstantinos Psounis

- Developed an internal knowledge-distillation method for transferring reasoning capabilities from larger teachers into sub-billion-parameter language models
- Replaced learned projection layers with fixed randomized transforms that approximately preserve pairwise distances between teacher representations
- Implemented a dynamic multi-objective loss combining language modeling, output-distribution distillation, and internal-representation alignment
- Evaluated two model families across two reasoning datasets and four benchmarks, reducing trainable parameters by up to 8.1% while keeping average scores within 0.15 percentage points of learned-projector baselines

KDFP: Knowledge Distillation from First Principles Aug 2025–Mar 2026
Collaborators: Konstantinos Psounis

- Developed a white-box LLM distillation method combining supervision of model outputs, embeddings, attention outputs, MLP outputs, and residual activations
- Implemented and evaluated over 100 distillation variants, investigating projection strategies, divergence objectives, and internal knowledge transfer
- Consolidated internal activation alignment into a shared learned projection matrix and evaluated teacher size, teacher fine-tuning, and training-data scaling

CPU-Auth: Hardware Authentication via Side-Channel Analysis Mar 2023–Jun 2024
Collaborators: Hari Venugopalan, Kaustav Goswami, Kartik Patwari, Chen-Nee Chuah, Zubair Shafiq

- Developed Siamese time-series classification pipelines for device authentication using browser-observed CPU dynamic voltage and frequency scaling (DVFS) fingerprints
- Benchmarked MLP, convolutional, and residual networks against dynamic time warping, achieving 95% validation accuracy with an MLP on held-out devices

TECHNICAL SKILLS

- **Programming Languages**: Python, C, C++, Java, Bash, Lua, Nix
- **Agentic AI**: Large Language Models (LLMs), AI Agents, LangGraph, Agent Harness Development, Workflow Orchestration, Model Context Protocol (MCP), Tool Integration, Knowledge Base Integration
- **Machine Learning Frameworks**: PyTorch, TensorFlow, Hugging Face Transformers, PyTorch Lightning, scikit-learn, Ray
- **LLM Training & Evaluation**: Supervised Fine-Tuning (SFT), Parameter-Efficient Fine-Tuning (PEFT), Low-Rank Adaptation (LoRA), Knowledge Distillation, LLM Evaluation, LLM-as-a-Judge, Prompt Injection Testing
- **Interpretability & Representation Analysis**: Sparse Autoencoders (SAEs), TransformerLens, SAELens, Representational Similarity Analysis (RSA), Centered Kernel Alignment (CKA)
- **Scientific Computing & Data Analysis**: NumPy, pandas, SciPy
- **Inference & Computing**: vLLM, Linux, High-Performance Computing (HPC)
- **Developer Tools**: Git, Docker, Nix, Weights & Biases (W&B)

SELECTED PROJECTS

Fully Reproducible System and Home Configurations with Nix Dec 2025–Present
github.com/ryanaswift7/nix-dots

- Binary-reproducible configuration of NixOS system
- Portable Home Manager configuration works even on non-NixOS systems
- Hierarchical, dendritic organization

LoRA Fine-tuning Implementation Sep–Dec 2024
github.com/ryanaswift7/gpt2_LoRA_567

- Reproduced original LoRA paper results, then implemented 6.85% improvement over baseline